# Classification Matrics

a) Accuracy Score:

(i) Binary Classification. [either 0 or 1 as output]

| | | | | | |
|---|---|---|---|---|---|
| label | 1 | 0 | 0 | 1 | 1 |
| Prediction | 0 | 0 | 0 | 1 | 1. |
| | ✗ | ✓ | ✓ | ✓ | ✓. |

$$\text{accuracy_score} = \frac{\text{total correct predictions}}{\text{total no. of predictions}}$$

$$= \frac{4}{5}$$

$$= 0.8 \rightarrow 80\%$$

(ii) Multi-Classification problem. [0, 1, 2, 3].

| | | | | | | |
|---|---|---|---|---|---|---|
| Actual label | 0 | 1 | 2 | 3 | 0 | 1 |
| Prediction | 0 | 1 | 2 | 3 | 1 | 0 |
| | ✓ | ✓ | ✓ | ✓ | ✗ | ✗ |

Again: $\text{accuracy_score} = \frac{4}{6}$

$$= 0.667 \rightarrow 66.67\%$$

(Q) How much accuracy is good?

→ It depends upon the problem that solving.

(a) Heart X-Ray Prediction model with 99% accuracy is bad.

→ coz one X-Ray in 100 will be wrong & hospital won't buy that model from you coz let's say 100000 people come daily in hospital for X-Ray. Your model gets 1000 prediction wrong daily

(b) Customer churn model → If its ~~doe~~ accuracy will be 90%, then it is better.

# Problem with Accuracy Score.

Let's say 90% accuracy.
Then it is doing 10% mistake. But, it doesn't tell us what kind of mistake has done by model.

Like, placement:

a) Yes $\xrightarrow{\text{But}}$ No
actual model

b) No $\xrightarrow{\text{But}}$ Yes
actual model

## b) Confusion Metrix

confusion_metrix(y_test, y_pred)

Output: array([26, 6],
[0, 29])

| | | Predicted (1) | Predicted (0) |
|---|---|---|---|
| Actual | (1) | 26 | 6 |
| | (0) | 0 | 29 |

26 & 29 corrects.

26 → correct prediction of heart dis
6 → model no, but actual yes.
0 → model yes, real no.
29 → model no, real no.

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Prediction

| | | 1 | 0 |
|---|---|---|---|
| Actual. | 1 | True Positive (TP) | False Negative (FN) |
| | 0 | False Positive (FP) | True Negative (TN) |

$$\text{accuracy_score} = \frac{TP + TN}{TP + FN + FP + TN}$$

# Type 1 Error = no. of false positive
# Type 2 Error = no. of ~~true~~ false negative.

Prediction

| | | 0 | 1 | 2 |
|---|---|---|---|---|
| Actual | 0 | 17 | 0 | 5 |
| ~~Prediction~~ | 1 | 2 | 21 | 6 |
| | 2 | 9 | 1 | 13 |

3×3.

$$\text{accuracy_score} = \frac{\text{true prediction}}{\text{total prediction}}$$